

Binance BTCUSDT Historical Data Downloader & MultiCharts Converter

Overview

This project automates the complete process of downloading, extracting, validating, merging, and converting Binance BTCUSDT 1-minute historical market data into a format suitable for MultiCharts.

The script downloads all available monthly BTCUSDT Spot 1-minute archives from Binance Vision, extracts the raw CSV files, combines them into a single dataset, performs multiple validation and cleanup steps, resolves known data anomalies, and produces a final CSV file ready for import into MultiCharts.

In addition to standard OHLCV data, the script also exports aggressive buyer volume, calculated seller volume, and trade count statistics, making the resulting dataset suitable for order flow analysis, volume studies, and advanced strategy development.

The primary goal of the project is to eliminate the manual work normally required to maintain a complete BTCUSDT historical database while ensuring data consistency and compatibility with trading platforms.

Directory Structure

Default location:

```
C:\binance_data\BTCUSDT
|
├── zip
│   ├── BTCUSDT-1m-2017-08.zip
│   ├── BTCUSDT-1m-2017-09.zip
│   └── ...
├── csv
│   ├── BTCUSDT-1m-2017-08.csv
│   ├── BTCUSDT-1m-2017-09.csv
│   └── ...
└── BTCUSDT_1m_ALL.csv
```

The script can also be executed with a custom data directory supplied through the command line.

Example:

```
python binance_btcusdt_all.py "D:\MarketData\BTCUSDT"
```

Workflow

Step 1 - Generate Download List

The script automatically generates a list of monthly BTCUSDT 1-minute archives.

Start date:

2017-08

End date:

Last completed month

The current month is intentionally excluded because Binance may still be updating its contents.

Step 2 - Download Missing Archives

Each archive is downloaded from Binance Vision.

Before downloading, the script checks whether the file already exists.

If the archive is already present, it is skipped.

Benefits:

- Fast incremental updates
 - Reduced bandwidth usage
 - Safe re-execution of the script
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Step 3 - Extract ZIP Archives

All downloaded archives are extracted into the CSV directory.

The script tracks:

- Total ZIP archives found
- Successfully extracted archives
- Skipped archives

- Extraction errors

Archives that have already been extracted are skipped automatically.

Step 4 - Load Historical Data

All extracted CSV files are loaded into memory.

For every file, the script reads the following Binance fields:

```
open_time
open
high
low
close
volume
trades
taker_buy_base
```

Additional Binance fields that are not required for the final export are ignored.

The script records:

- Total CSV files found
 - Successfully loaded files
 - Failed files
 - Total rows loaded
-

Step 5 - Timestamp Validation

Binance historical datasets are not always perfectly uniform.

The script validates all timestamps before processing.

Header Detection

Occasionally a CSV file may contain an unexpected header row.

The script converts timestamps to numeric values and removes invalid rows.

Reported as:

```
Header rows removed
```

Timestamp Range Validation

Only timestamps within the expected historical range are accepted.

Rows outside the valid range are removed.

Reported as:

Out-of-range rows removed

Microsecond Timestamp Detection

Some datasets may contain timestamps stored in microseconds instead of milliseconds.

Example:

1512367200000000

instead of

1512367200000

The script automatically detects and converts such values.

Reported as:

Microsecond timestamps detected

This prevents corrupted dates and duplicate bars caused by inconsistent timestamp precision.

Step 6 - Date and Time Conversion

After validation, Binance timestamps are converted into human-readable format:

MM/DD/YYYY
HH:MM

Example:

12/04/2017
06:00

This format is directly compatible with MultiCharts ASCII Import.

Step 6.1 - Volume Decomposition

The script derives additional market activity metrics from Binance trade statistics.

Up Volume

Represents volume initiated by aggressive buyers.

Source:

taker_buy_base

Down Volume

Calculated as:

Down Volume = Volume - Up Volume

This represents volume initiated by aggressive sellers.

Total Ticks

Represents the number of individual trades executed during the bar.

Source:

trades

These additional fields are useful for:

- Order flow analysis
 - Volume imbalance studies
 - Delta-style indicators
 - Trade activity analysis
 - MultiCharts custom indicators and strategies
-

Step 7 - Duplicate Minute Bar Resolution

A known issue in some Binance historical datasets is the presence of multiple records that map to the same minute after conversion.

Example:

```
open_time=1512367200000
Date=12/04/2017
Time=06:00
Volume=0.289485
```

```
open_time=1512367220799
Date=12/04/2017
Time=06:00
Volume=0.000000
```

Although these timestamps are different, they both belong to the same minute.

MultiCharts cannot import duplicate minute bars.

To resolve this:

1. Duplicate Date+Time bars are detected.
2. Bars are sorted by volume.
3. The bar with the highest volume is preserved.
4. Lower-volume duplicates are removed.

This preserves the most meaningful market data while maintaining MultiCharts compatibility.

Step 8 - Export Final Dataset

The final dataset is written to:

```
BTCUSDT_1m_ALL.csv
```

Columns:

```
Date
Time
Open
High
Low
Close
Volume
Up Volume
```

```
Down Volume
Total Ticks
```

Field descriptions:

Column	Description
Date	Trading date (MM/DD/YYYY)
Time	Trading time (HH:MM)
Open	Opening price
High	Highest price
Low	Lowest price
Close	Closing price
Volume	Total traded volume
Up Volume	Aggressive buyer volume (Binance taker_buy_base)
Down Volume	Calculated seller volume (Volume - Up Volume)
Total Ticks	Number of trades reported by Binance

Example:

```
12/04/2017,06:00,11476.87,11478.00,11476.87,11478.00,0.289485,0.289485,0.000000,4
```

The file is overwritten on every execution to ensure the latest history is always available.

Debug Logging

The script provides extensive diagnostic output during execution.

Download Phase

Reports:

```
Files to download
Archive already exists
Archive downloaded successfully
Archive not found
Download errors
```

Extraction Phase

Reports:

```
ZIP archives found
Archives extracted
Archives skipped
Extraction errors
```

Loading Phase

Reports:

```
CSV files found
CSV files loaded
CSV files failed
Rows loaded
```

Data Validation Phase

Reports:

```
Header rows removed
Microsecond timestamps detected
Out-of-range rows removed
Rows after cleanup
```

Duplicate Detection Phase

Reports:

```
Duplicate open_time values
Duplicate Date+Time values
Duplicate minute bars removed
```

If duplicates are found, detailed information is printed including:

```
Timestamp
Date
```

Time
Open
High
Low
Close
Volume

This information is useful for investigating data anomalies originating from Binance historical archives.

Export Phase

Reports:

Rows before cleanup
Rows saved
Historical range
Output file location

Example:

Rows saved: 4,321,500

Range:
08/17/2017 04:00
-> 05/31/2026 23:59

Output file:
C:\binance_data\BTCUSDT\BTCUSDT_1m_ALL.csv

Intended Use Cases

This project is designed for:

- MultiCharts users
- TradeStation users
- Quantitative traders
- Historical market data collection
- Strategy development
- Backtesting
- Research and analytics
- Order flow analysis
- Volume imbalance studies

It is particularly useful for traders who require a continuously updated BTCUSDT historical database without manually downloading and processing Binance archives.

Key Features

- Automatic Binance archive discovery
- Incremental downloading
- Automatic ZIP extraction
- Historical data merging
- Timestamp normalization
- Microsecond-to-millisecond correction
- Data validation
- Duplicate minute bar resolution
- MultiCharts-compatible export
- Aggressive buyer volume extraction
- Aggressive seller volume calculation
- Trade count (tick count) export
- Detailed diagnostic logging
- Re-runnable without manual cleanup
- Custom data directory support
- Complete BTCUSDT 1-minute history generation with a single command